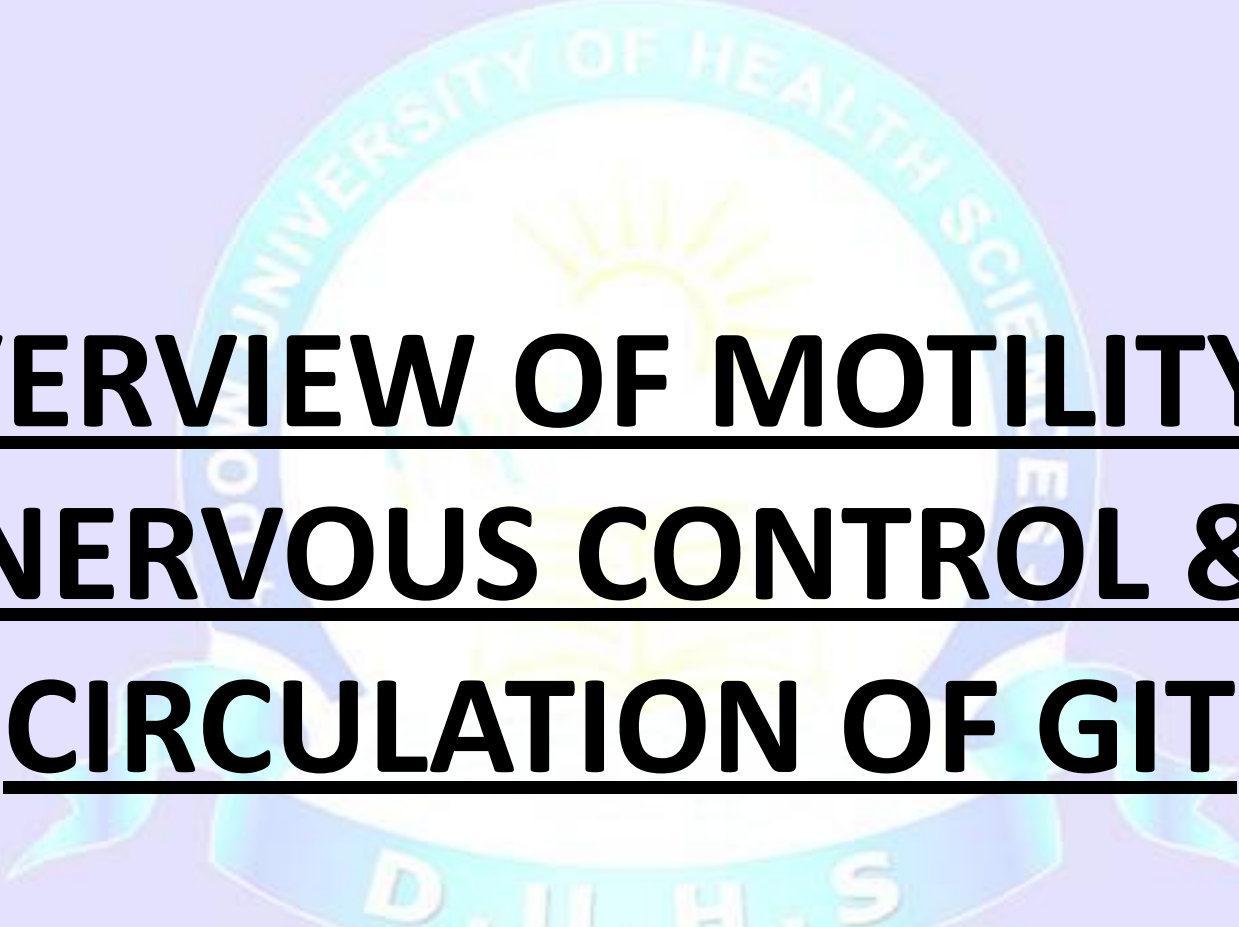
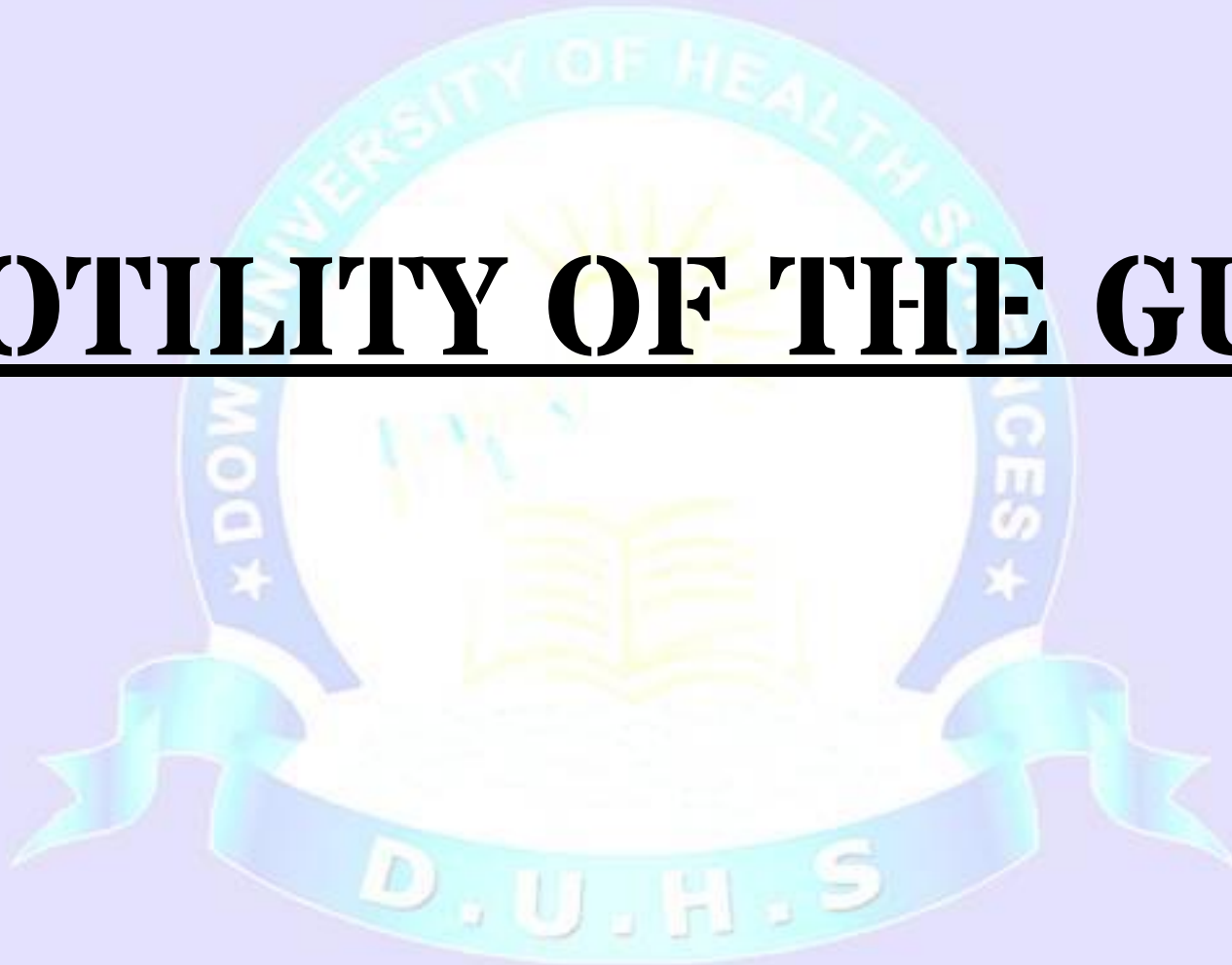


The logo of D.U.H.S. (Datta University of Health Sciences) is a circular emblem. It features a central sun with rays. The outer ring of the emblem contains the text "DATTATRI UNIVERSITY OF HEALTH SCIENCES" in a semi-circle. Below the emblem, a blue ribbon banner displays the acronym "D.U.H.S." in white capital letters.

OVERVIEW OF MOTILITY & **NERVOUS CONTROL &** **CIRCULATION OF GIT**

MOTILITY OF THE GUT



NERVES

+

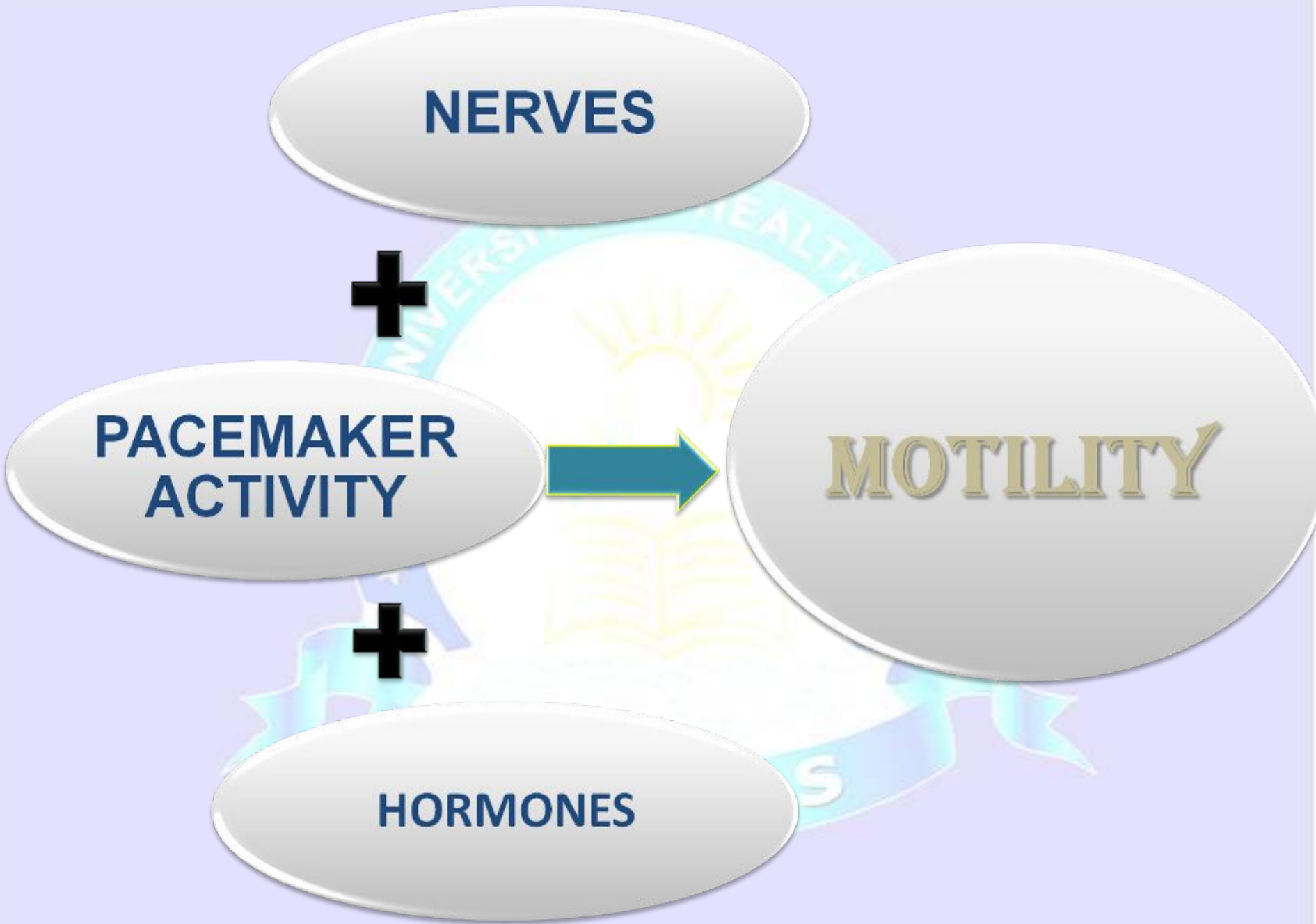
**PACEMAKER
ACTIVITY**

+

HORMONES

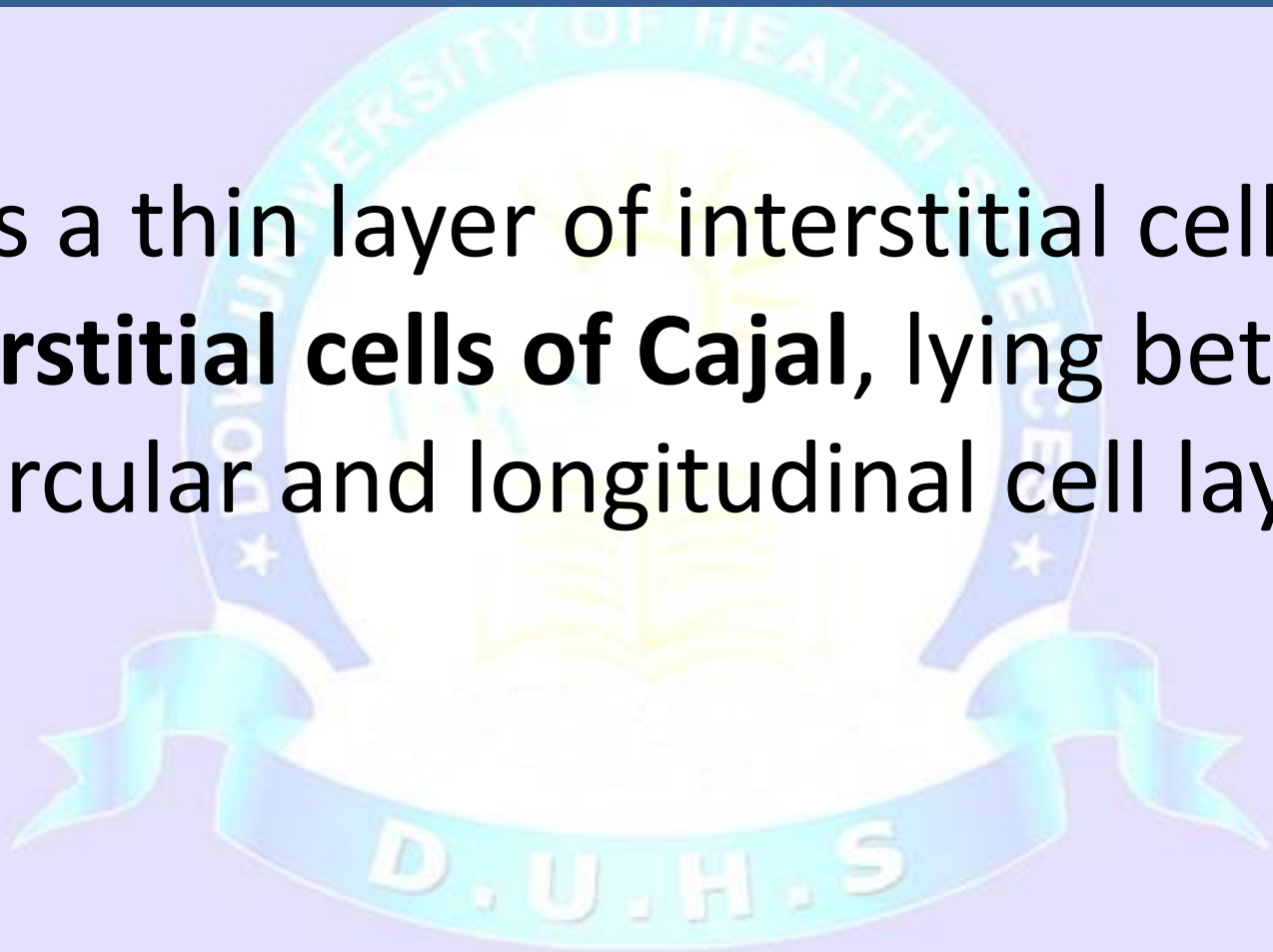


MOTILITY



PACEMAKER OF THE GUT

It is a thin layer of interstitial cells ;
interstitial cells of Cajal, lying between
circular and longitudinal cell layer.



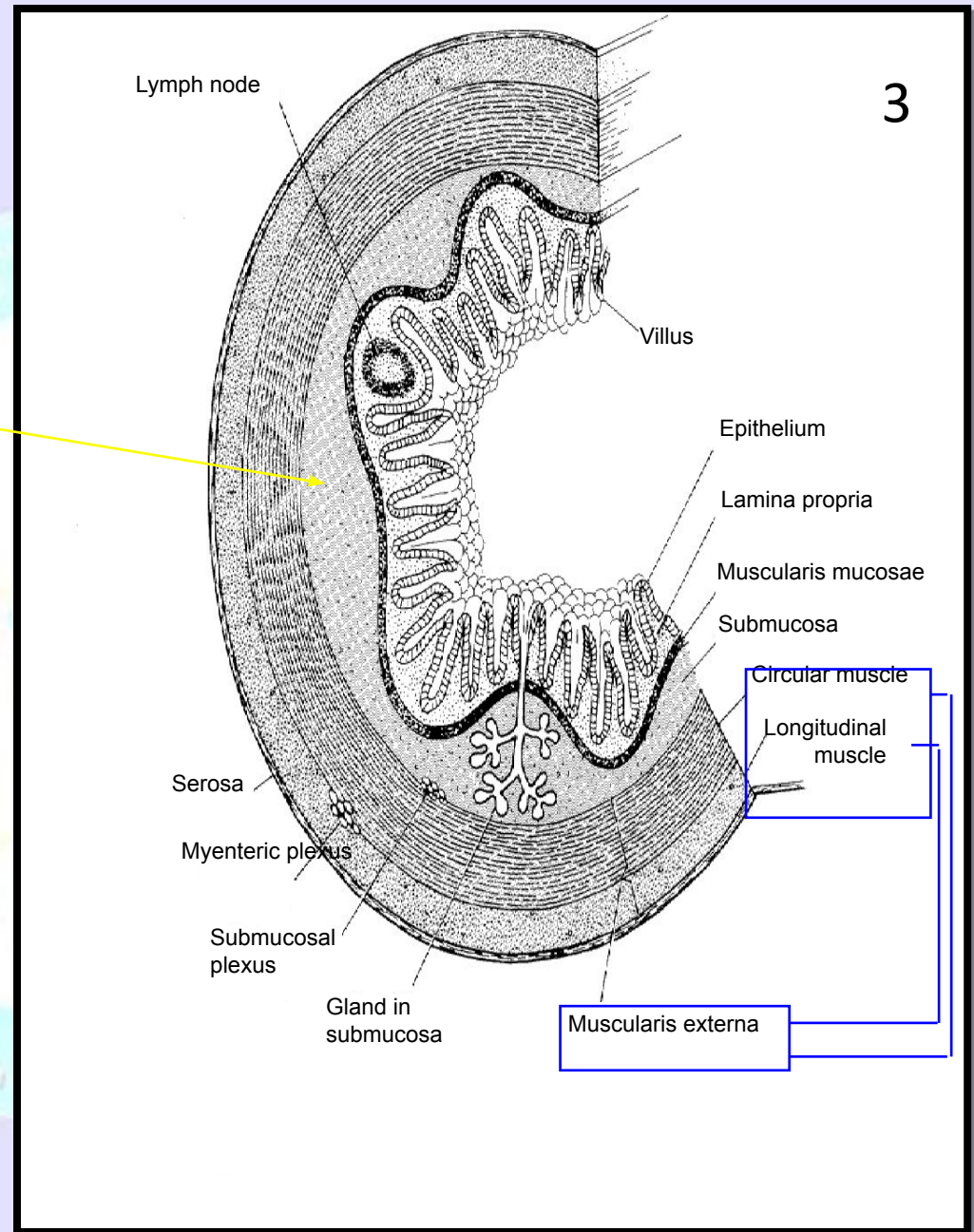
interstitial cells of Cajal

It is characterized by:

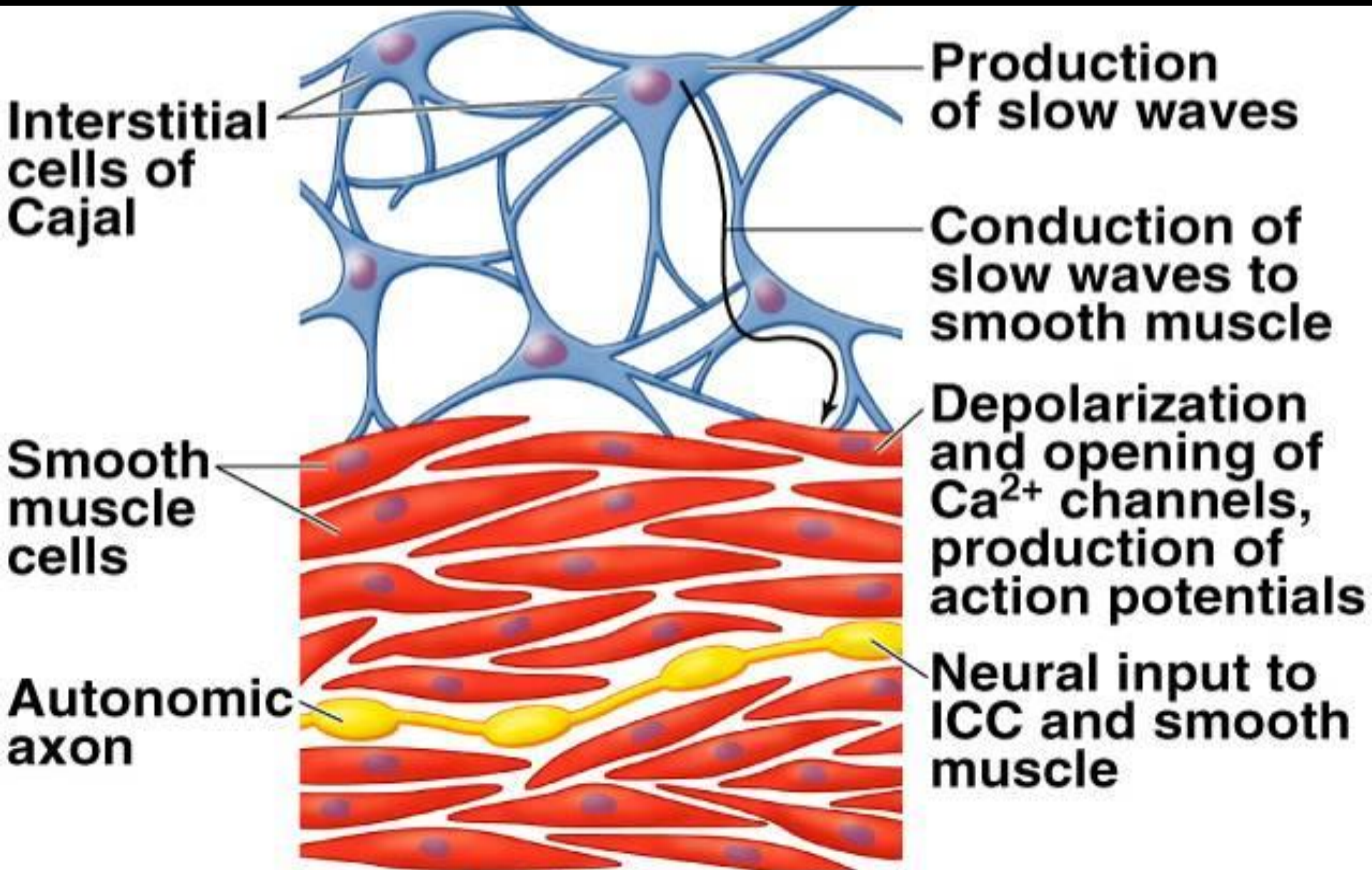
- ★ a slow wave activity
- ★ gap junctions
- ★ tight synapses with enteric nerves

Pacemaker activity:

Thin layer of
interstitial cells
(**interstitial cells
of Cajal**)
between circular
and longitudinal
cell layer.
Conduction
through gap
junctions.



PACEMAKER OF THE GUT



Electro-physiology of smooth muscles of the GUT



Regulation of gastric emptying

- **Gastric: promote emptying**

- Stretch mediated in response to volume
- Gastrin - promotes the pyloric pump

- **Duodenal: inhibit emptying**

- **Neural**

- Intrinsic
- Extrinsic

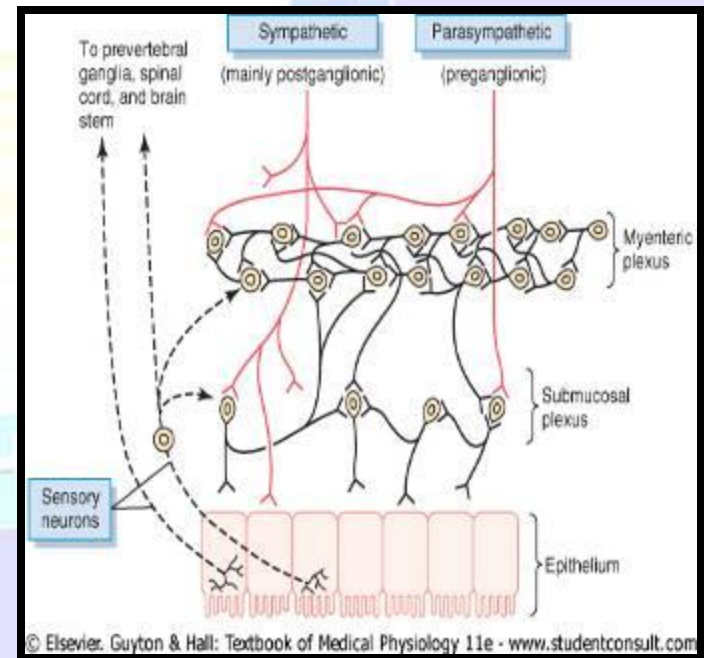
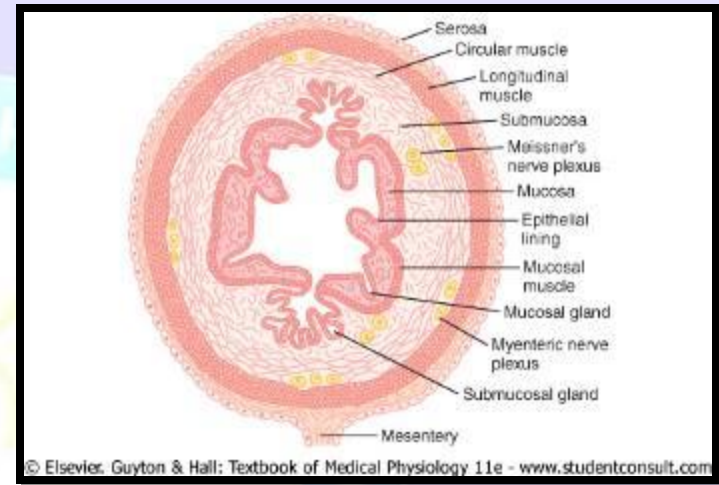
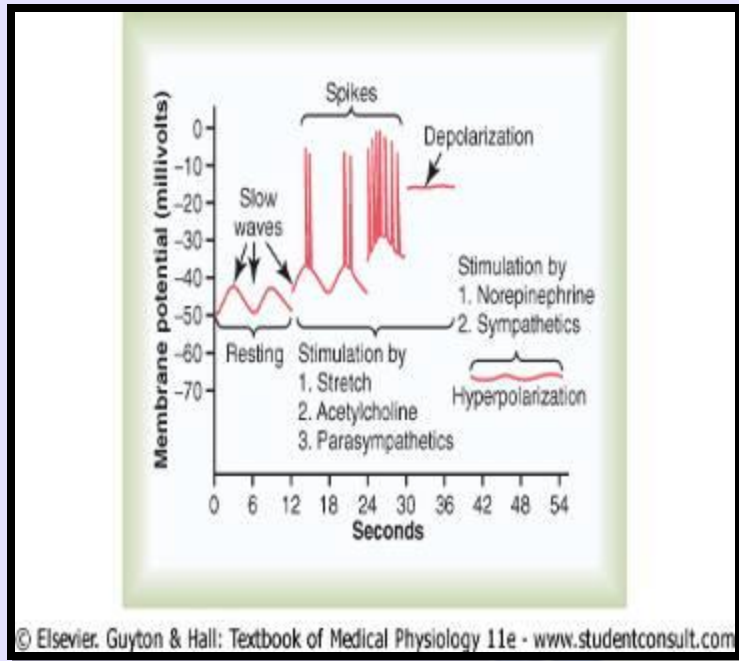
- **Endocrine**

- CCK
- Secretin
- GIP

} Distension, irritation- acidity/osmolar change

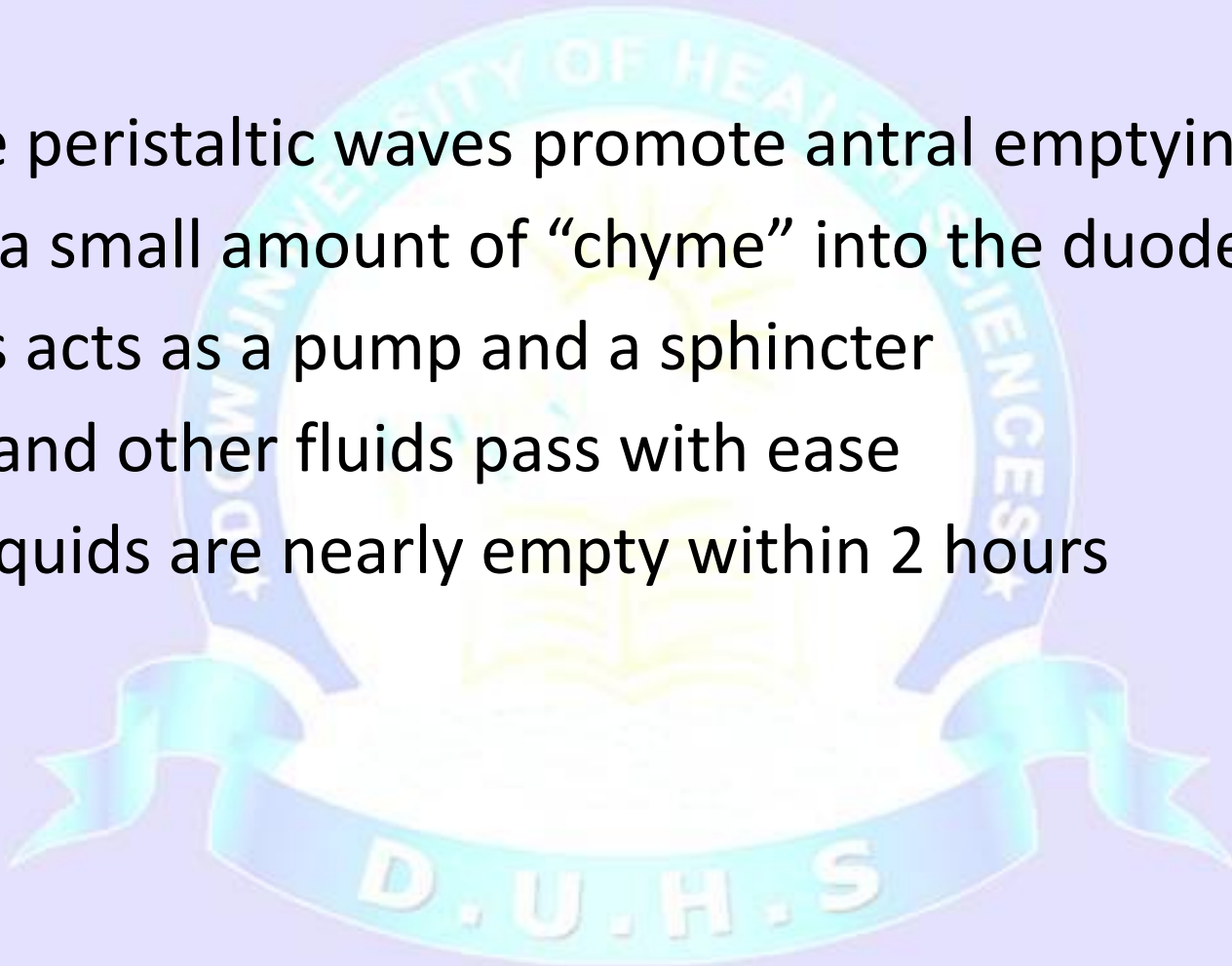
} Fats in chyme

Gastrointestinal motility

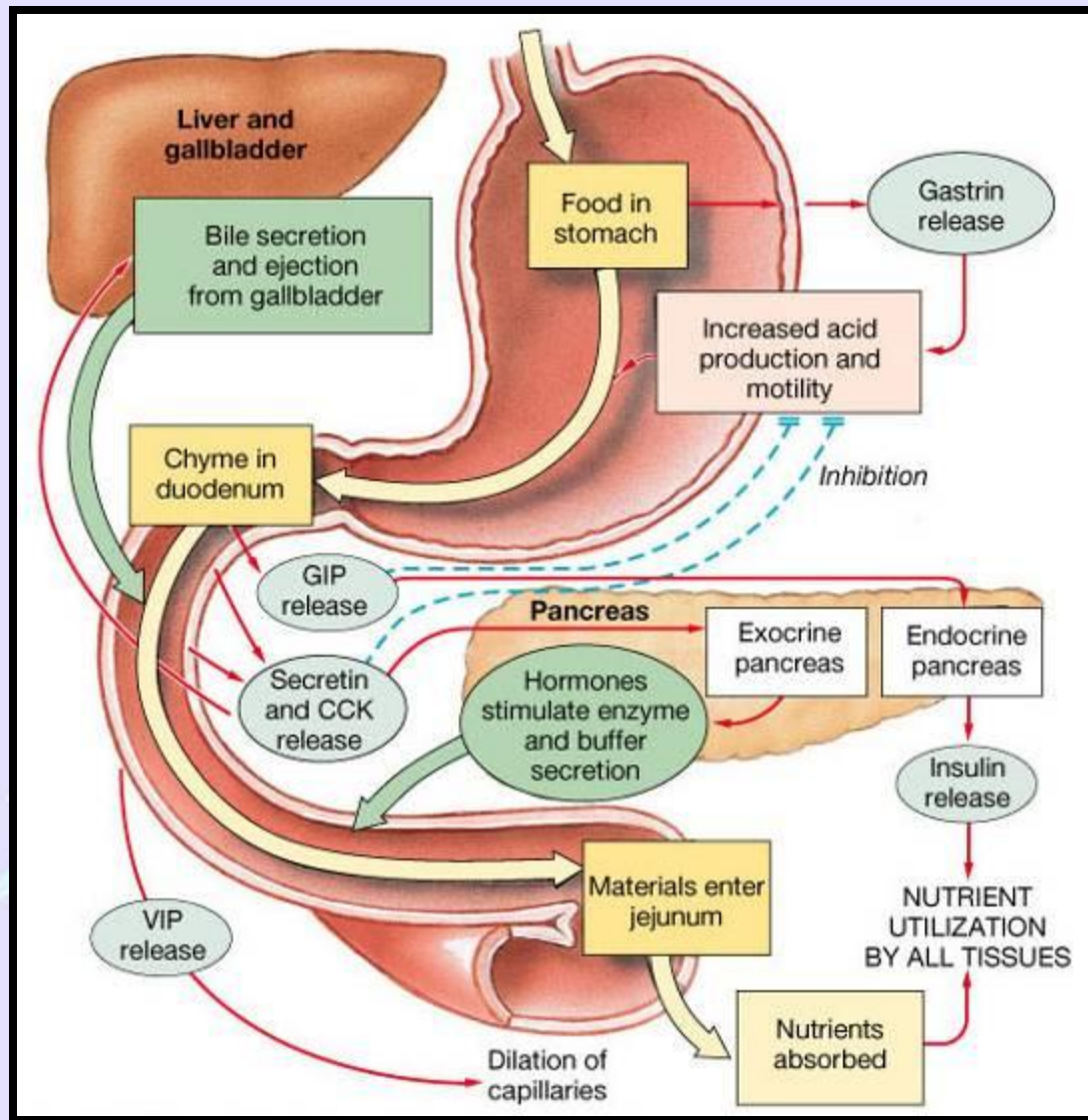


Gastric emptying

- Intense peristaltic waves promote antral emptying
- Forces a small amount of “chyme” into the duodenum
- Pylorus acts as a pump and a sphincter
- Water and other fluids pass with ease
- Clear liquids are nearly empty within 2 hours

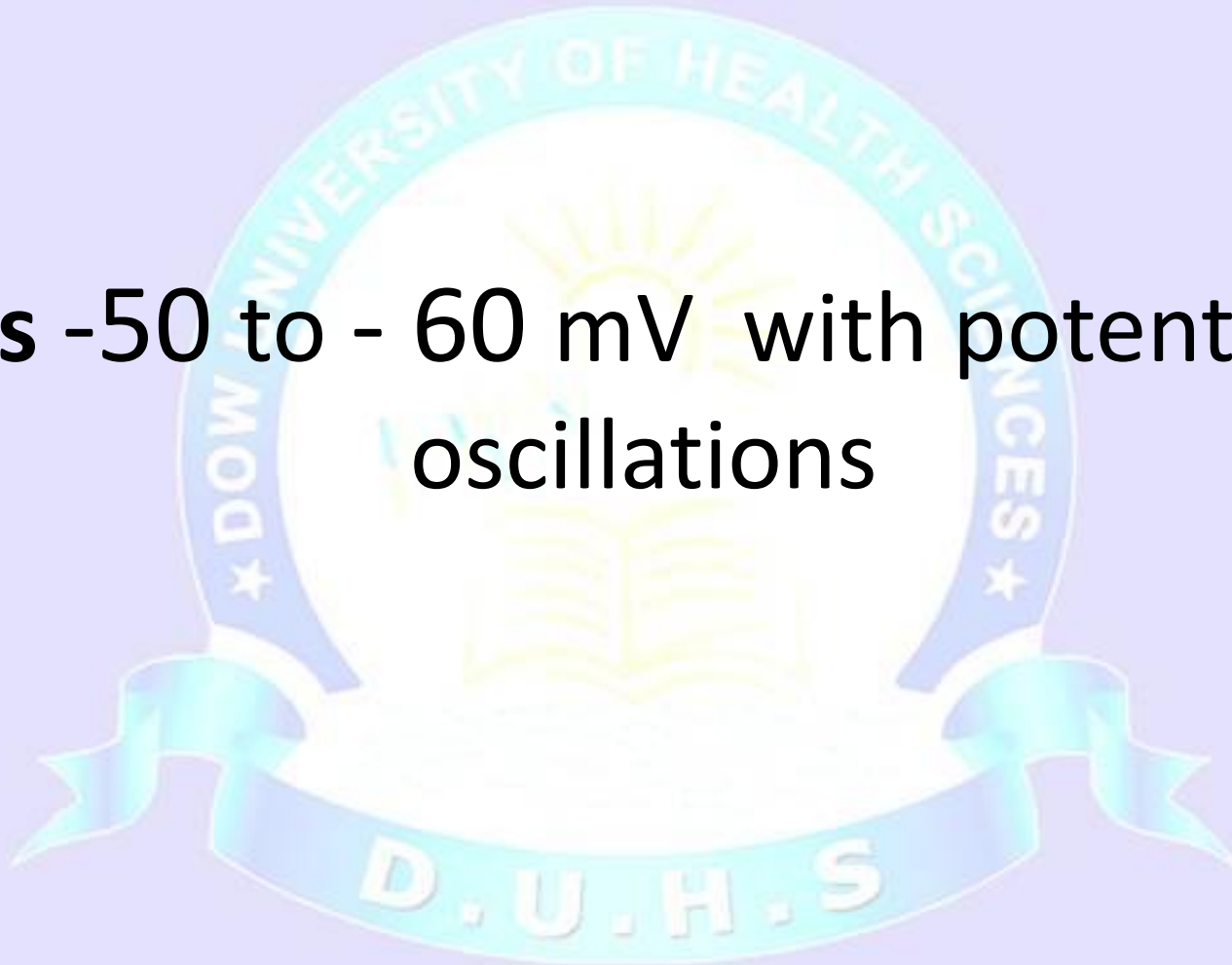


Overview of gastric emptying



Resting membrane potential

It is -50 to -60 mV with potential oscillations



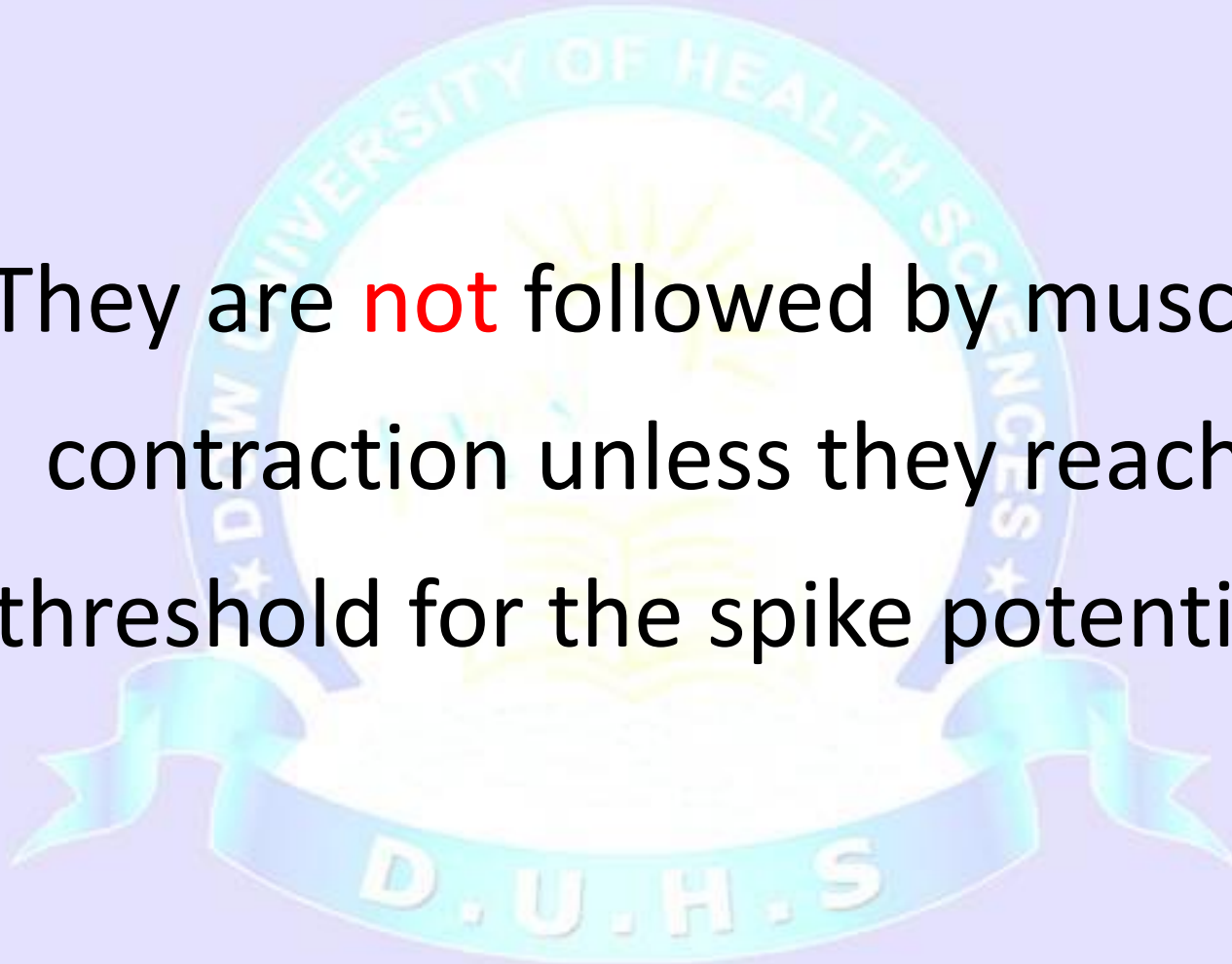
Slow Waves = Pacemaker Potential = Basic
Electric Rhythm (BER)

**It is spontaneous rhythmic fluctuation in
the smooth muscles membrane potential
between – 60 mv & – 40 mv.**

It is initiated by the interstitial cells of Cajal

Pacemaker potential

They are **not** followed by muscle contraction unless they reach a threshold for the spike potentials.

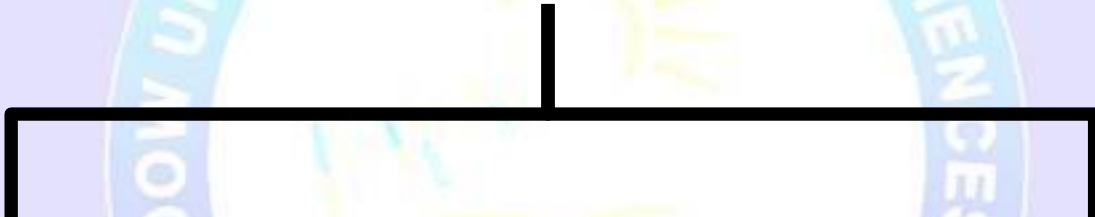


The action potentials (spike potentials)

is produced when slow-waves reach the threshold automatically due to **Opening of L-type calcium channels**

This is induced by **Stretch of smooth muscle, vagal stimulation, hormones or myenteric plexus stimulation.**

Functional types of movements

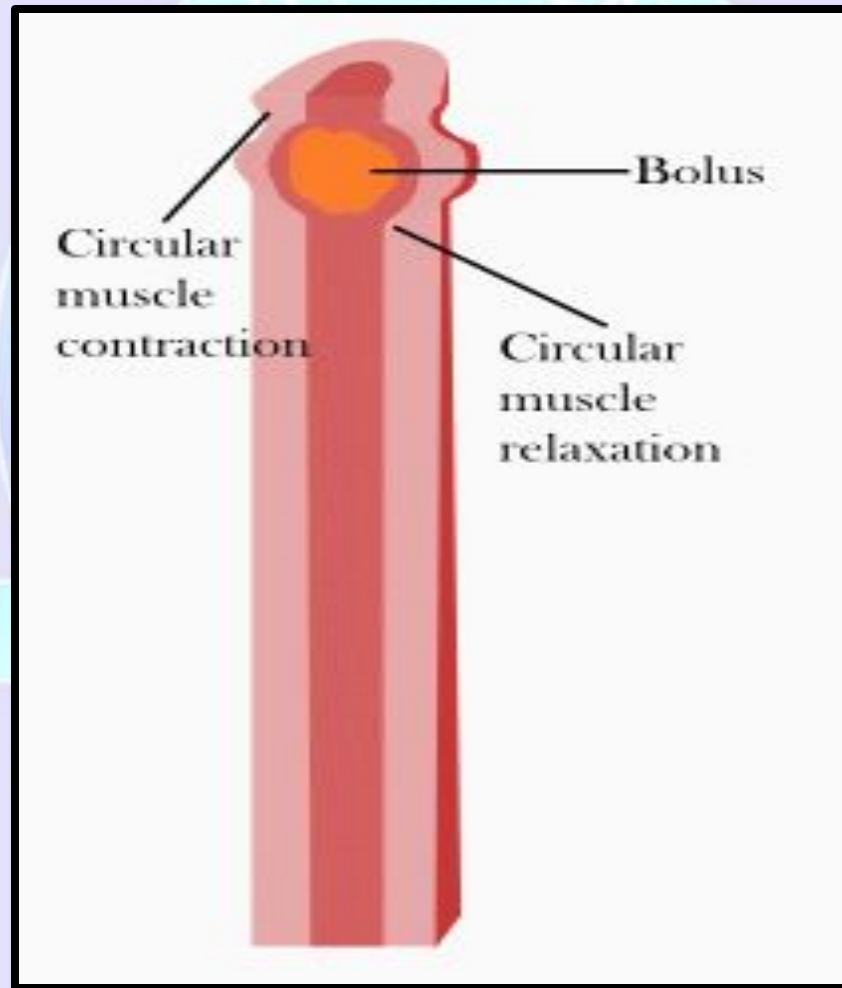


```
graph TD; A[ ] --- B[Propulsive movements (peristalsis)]; A --- C[Mixing movements];
```

**Propulsive
movements
(peristalsis)**

**Mixing
movements**

Peristalsis



Peristalsis

This signal is transmitted by
interneurons

Excitatory (proximal) motor neuron

Inhibitory (distal) motor neurons,

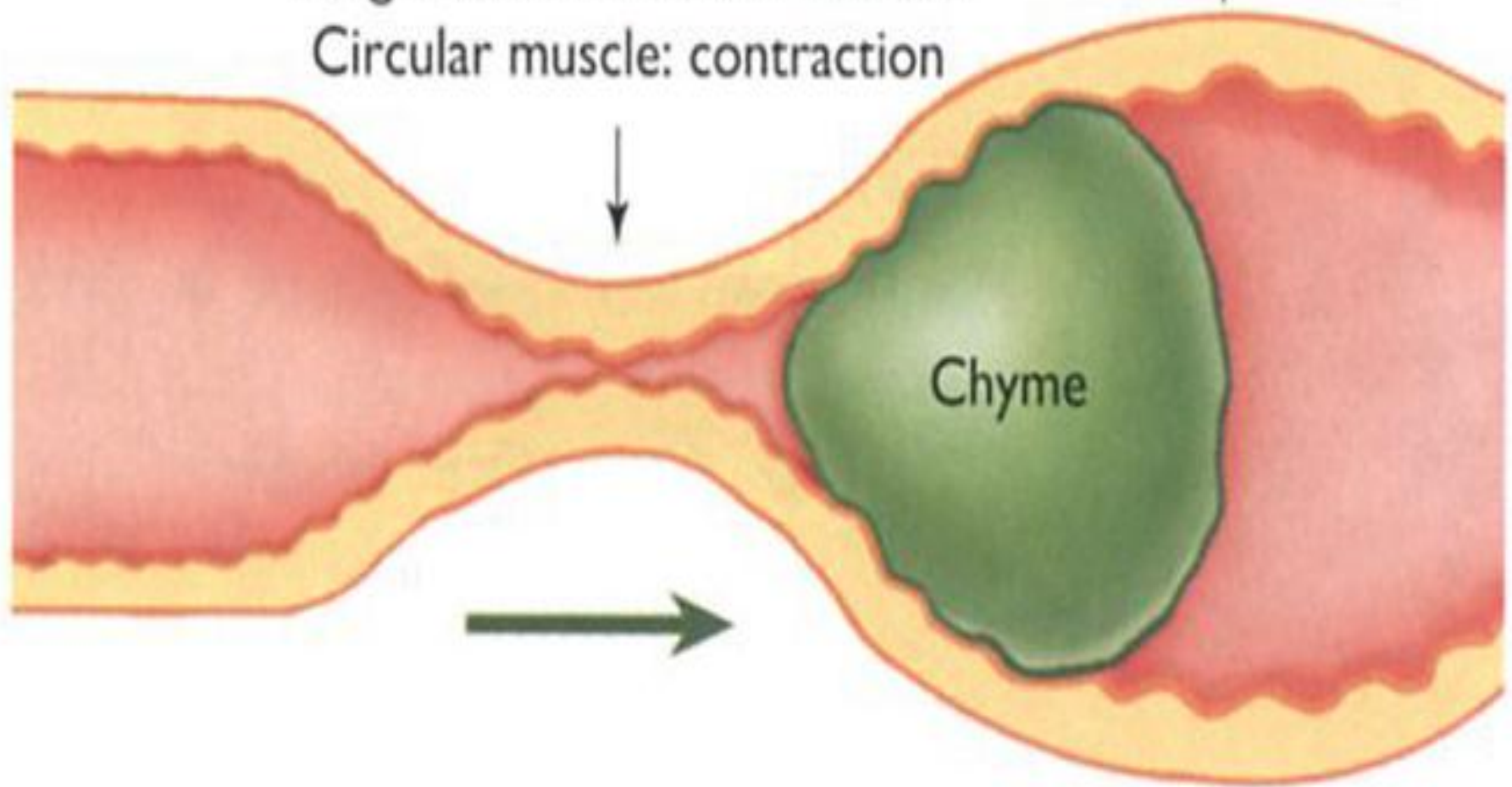
bolus is transported in oral to caudal
direction

Longitudinal muscle: contraction

Circular muscle: relaxation

Longitudinal muscle: relaxation

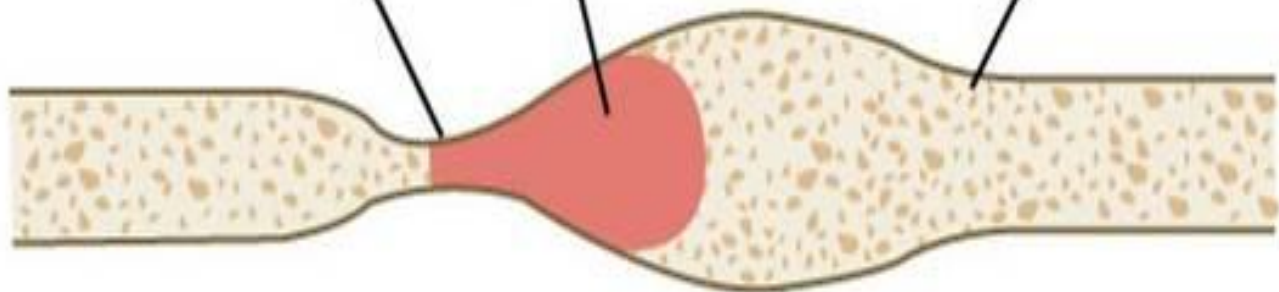
Circular muscle: contraction



**Peristaltic contractions are responsible
for forward movement**

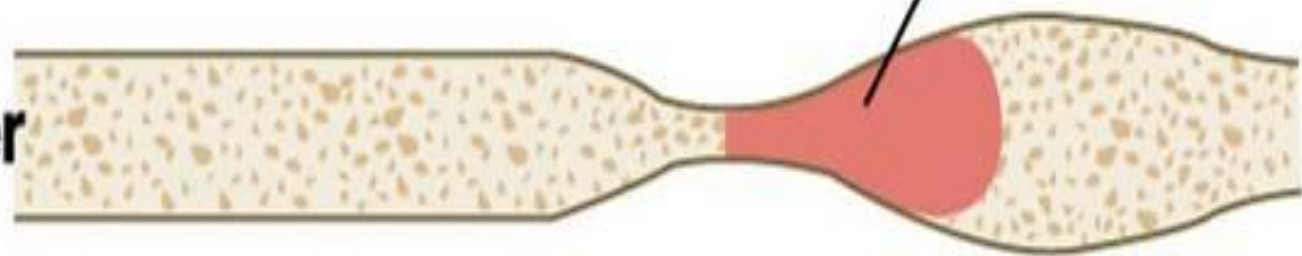
Contraction Bolus Receiving segment

Time zero



Bolus moves forward

Seconds later



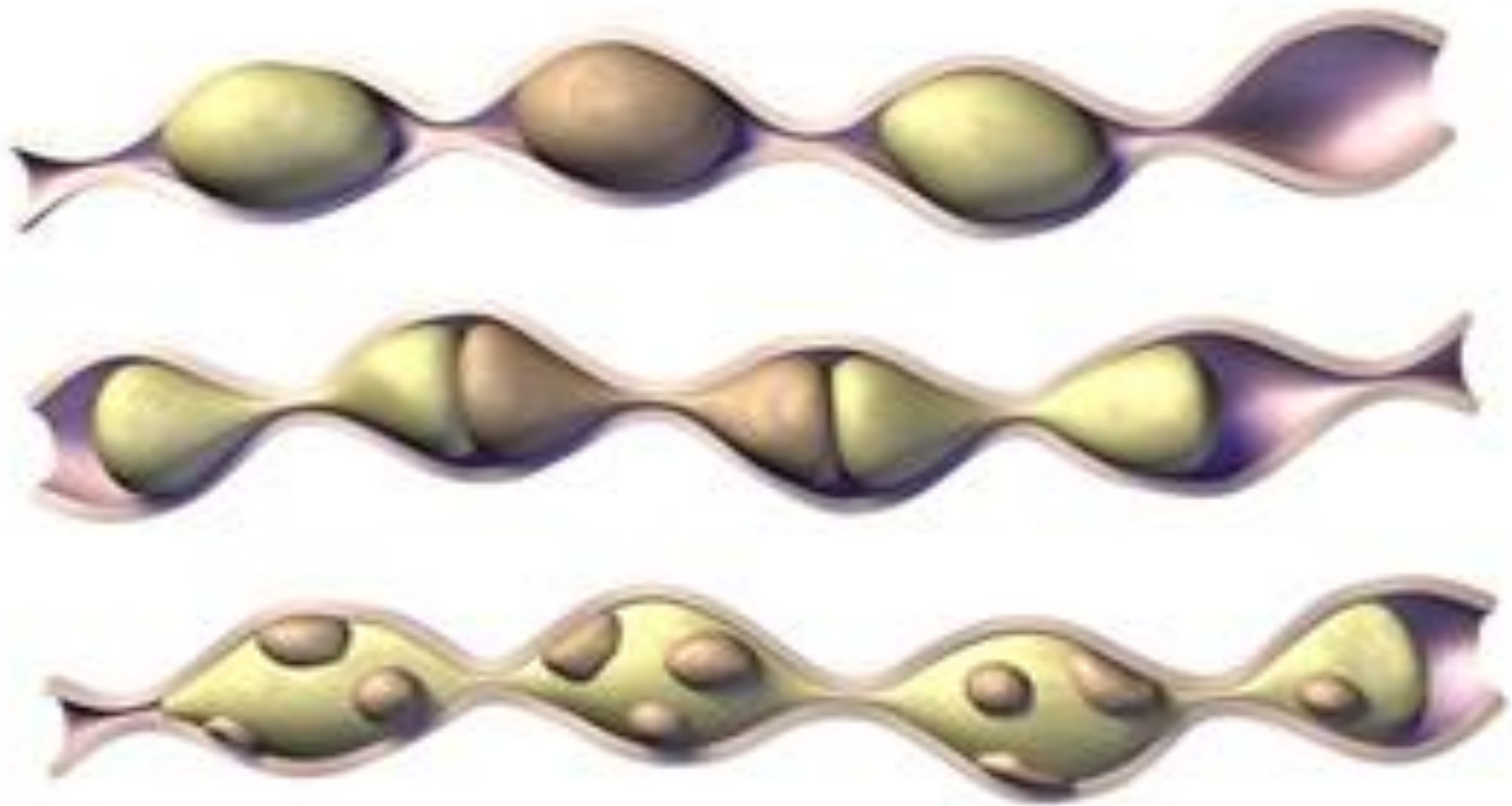
Direction of movement



Mixing movement

They are local intermittent constrictive contractions occur every few cms in the gut wall, lasts only 5 -30 seconds then new constrictions occur at other points keeping the contents thoroughly mixed together

Mixing movement



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Tonic contractions

continuous contractions with different intensity lasts minutes to hours.

It is caused by continuous repetitive spike potential, any factor causes continuous partial depolarization, or by hormones.

It is usually occurs at the sphincteric regions

•

Hunger contraction

It is rhythmical continuous tetanic intense prestalsis on empty stomach, lasts 2-3 minutes in young.

□ It may cause mild pain in the pit of stomach, hunger pangs especially after 12 hours – 4 days of starvation then it is gradually weakened.

Receptive relaxation of the **stomach:**

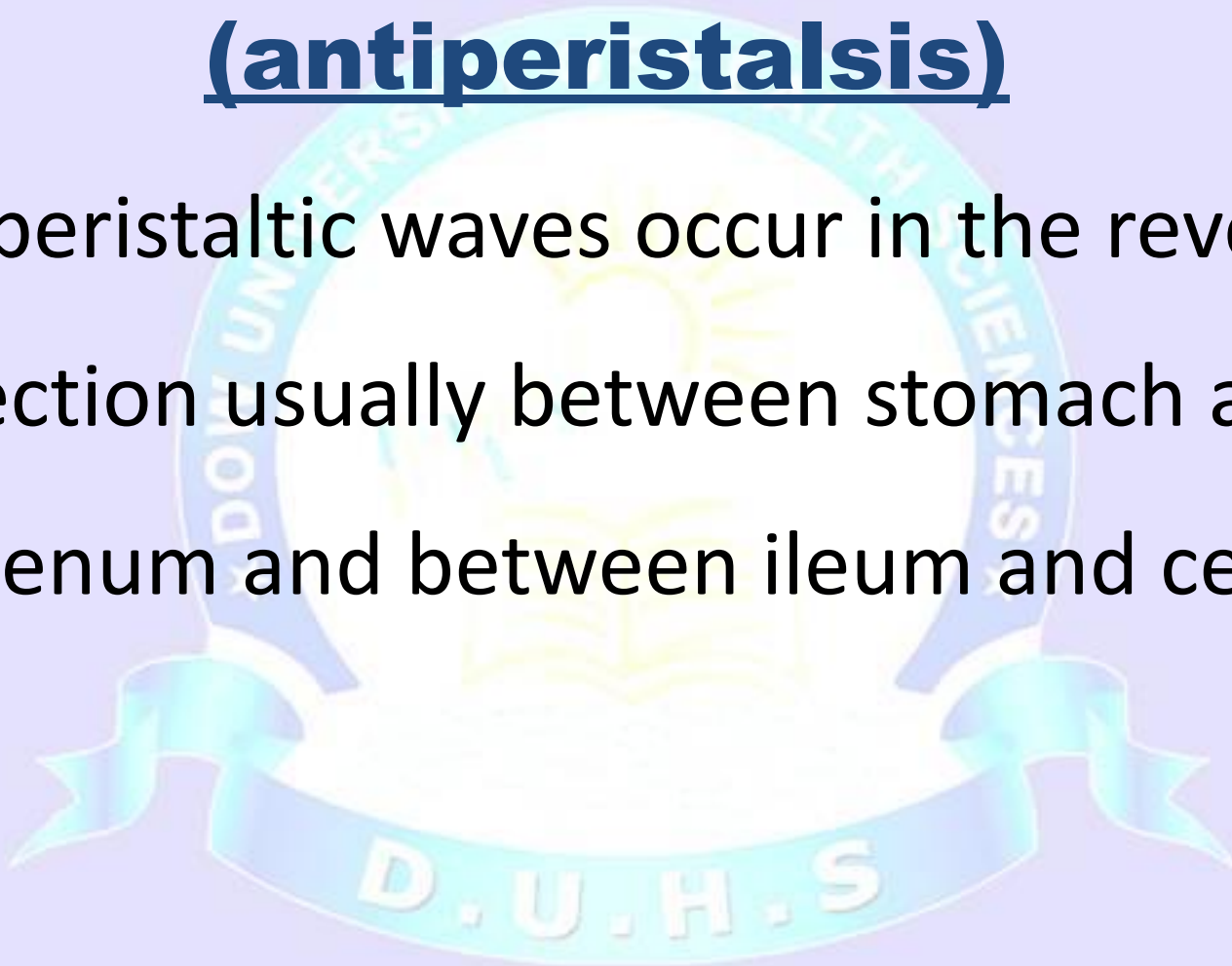
When the esophageal peristalsis approaches the stomach , a wave of relaxation occurs in the entire stomach to the duodenum to prepare to receive food.

It is due to ++ of myentric inhibitory neurons.

Reverse peristalsis

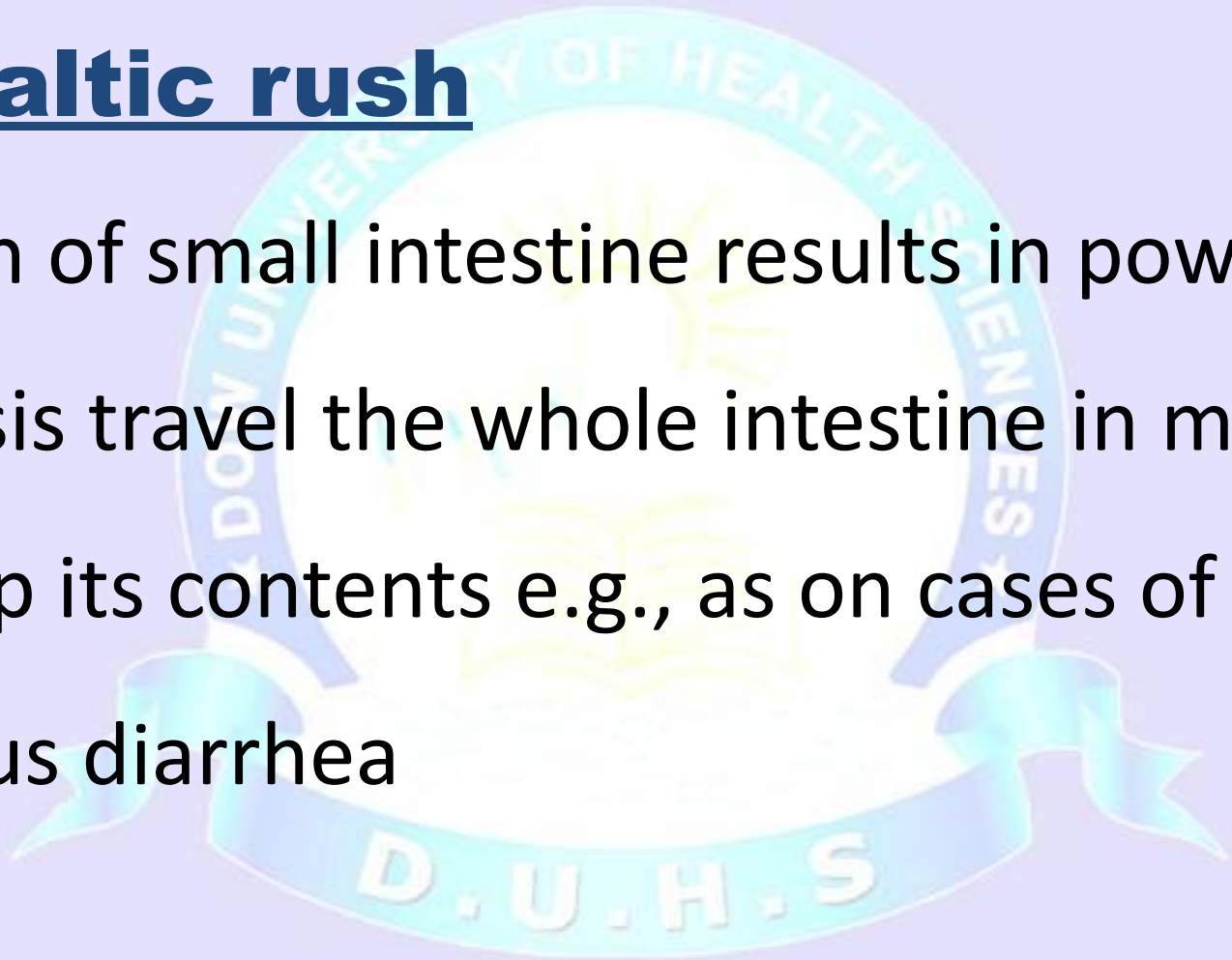
(antiperistalsis)

It is peristaltic waves occur in the reverse direction usually between stomach and duodenum and between ileum and cecum



Peristaltic rush

Irritation of small intestine results in powerful peristalsis travel the whole intestine in minutes to sweep its contents e.g., as on cases of infectious diarrhea



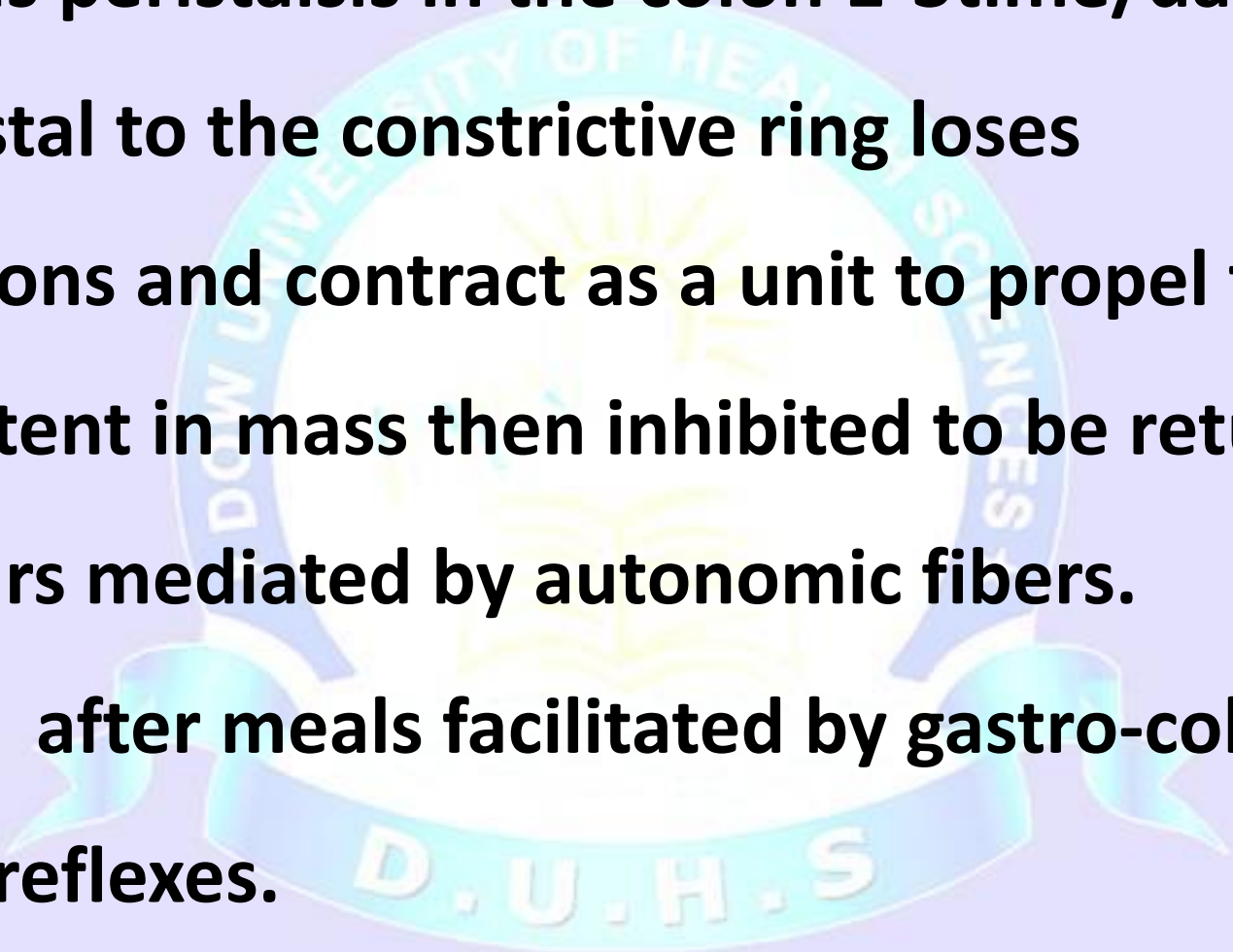
Haustration (segmentation of the colon)

it is simultaneous strong contraction of the circular muscles and teniae coli of the colon results in out plugging of the distal part to allow dug into and roll over the fecal material to absorb fluid and dissolved substances

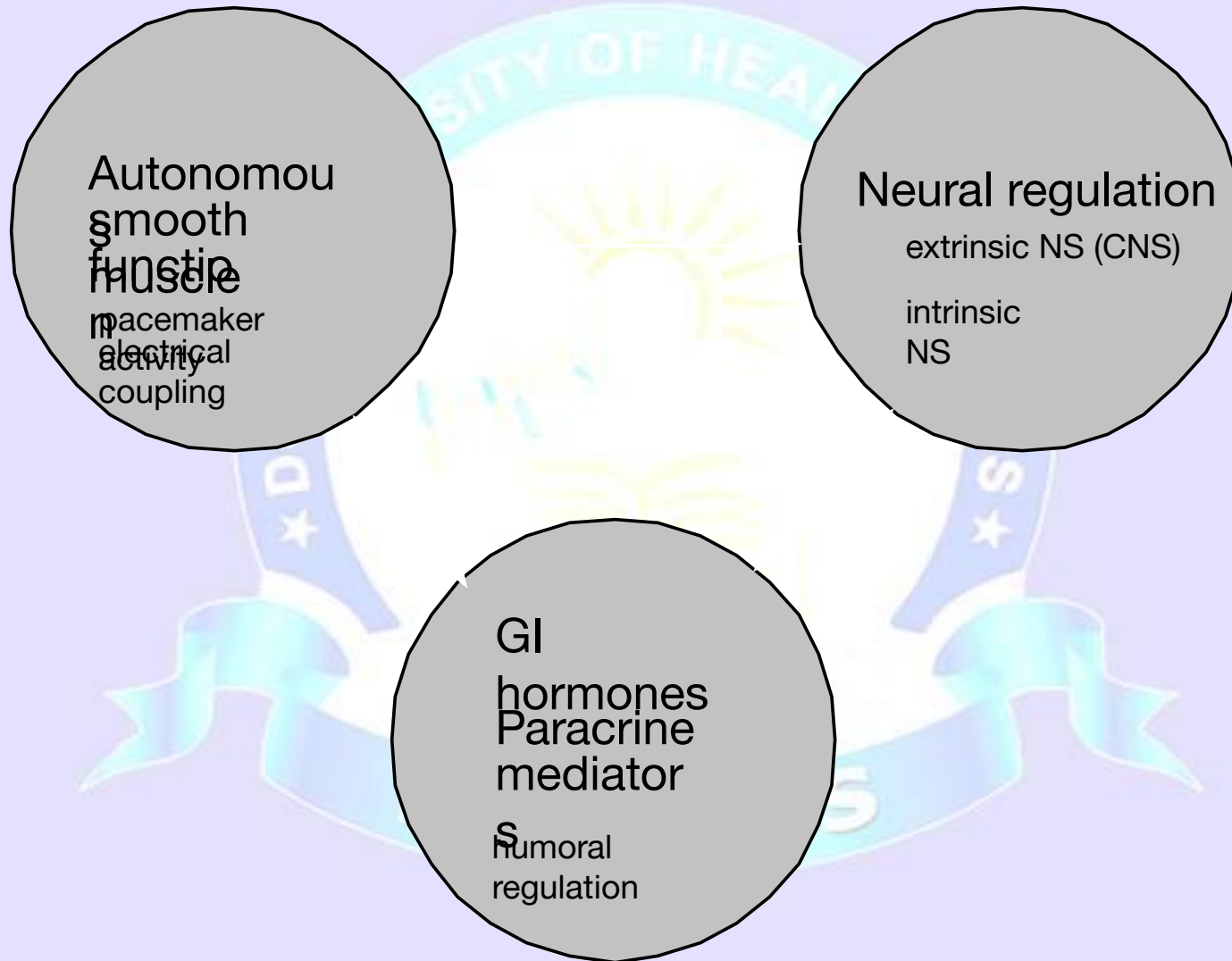
Mass movements

It is peristalsis in the colon 1-3time/day.

**20 cm distal to the constrictive ring loses
haustrations and contract as a unit to propel the
fecal content in mass then inhibited to be returned
after hours mediated by autonomic fibers.
It appear after meals facilitated by gastro-colic and
ilio-colic reflexes.**

The logo of Dow University of Health Sciences (D.U.H.S.) is a circular emblem. It features a central sunburst design with rays emanating from a central point. The words "DOW UNIVERSITY OF HEALTH SCIENCES" are written in a circular path around the sunburst. At the bottom of the emblem, the acronym "D.U.H.S." is displayed on a blue ribbon-like banner.

Regulation of GI function



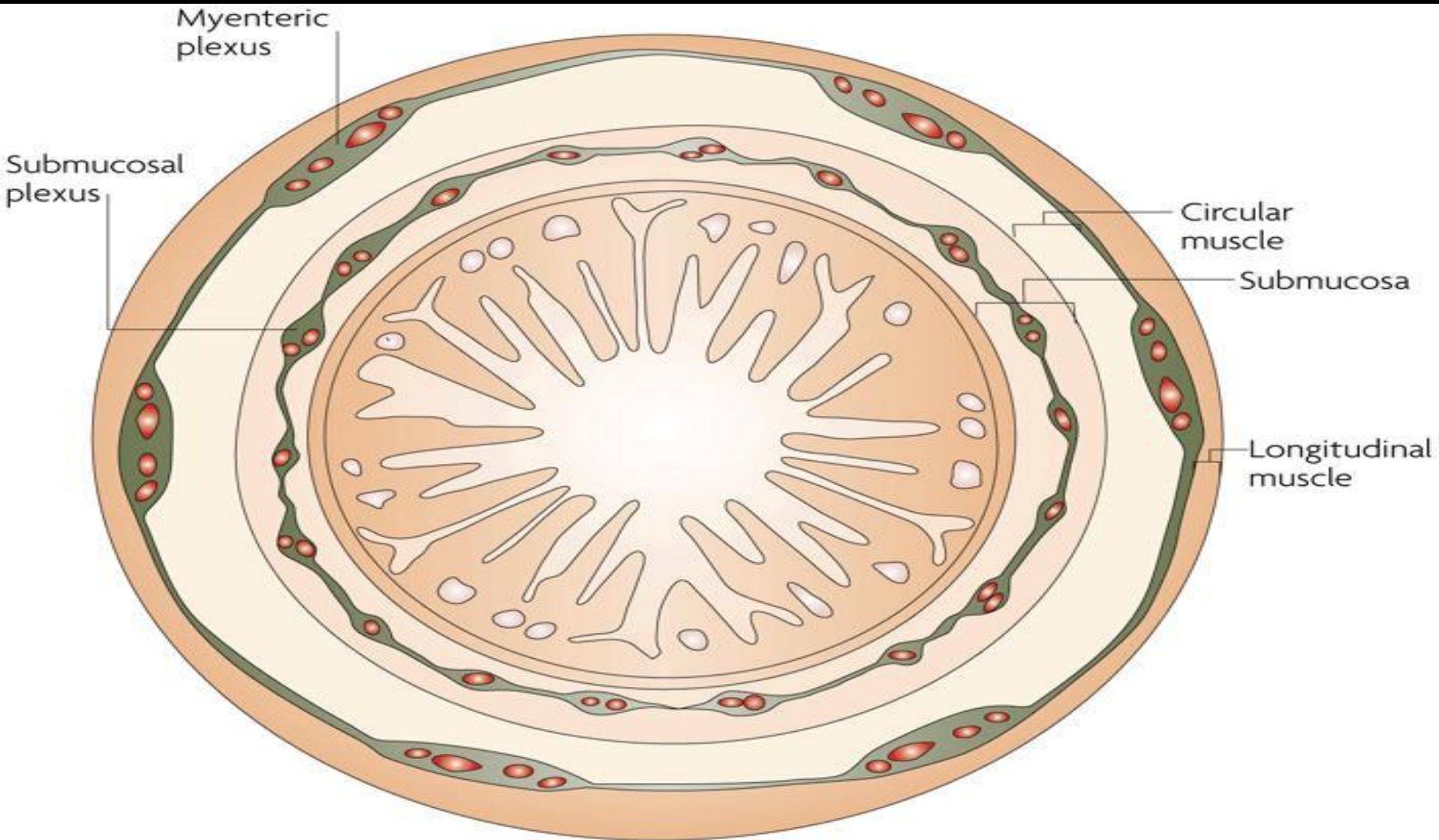
Regulation of functional components

Neural

- Enteric NS controls secretions and movement
- Comprised of 2 plexuses:
 1. Myenteric (Auerbach's)
 2. Submucosal (Meissner's)
- The plexuses are influenced by ANS

Hormonal

Enteric nervous system (gut brain)



Enteric nervous system (gut brain)

The myenteric plexus has

- ★ Sensory receptor neurons

- ★ Motor effector neurons

- ★ The neurotransmitters

Excitatory & Inhibitory

Nervous Control of the GI Tract

- Intrinsic controls
 - Nerve plexuses near the GI tract initiate short reflexes
 - Short reflexes are mediated by local enteric plexuses (gut brain)
- Extrinsic controls
 - Long reflexes arising within or outside the GI tract
 - CNS centers and extrinsic autonomic nerves

Nervous Control of the GI Tract

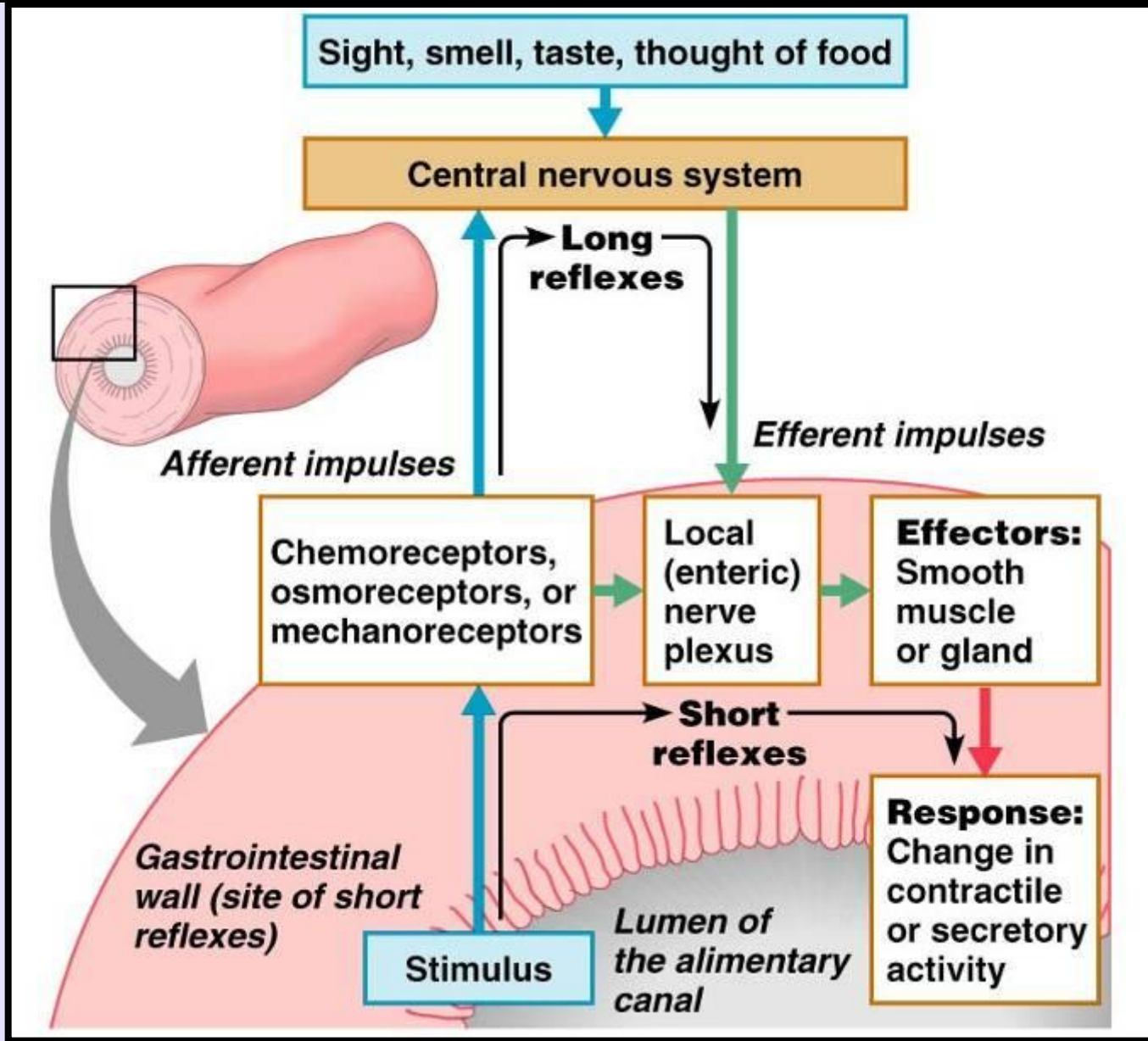
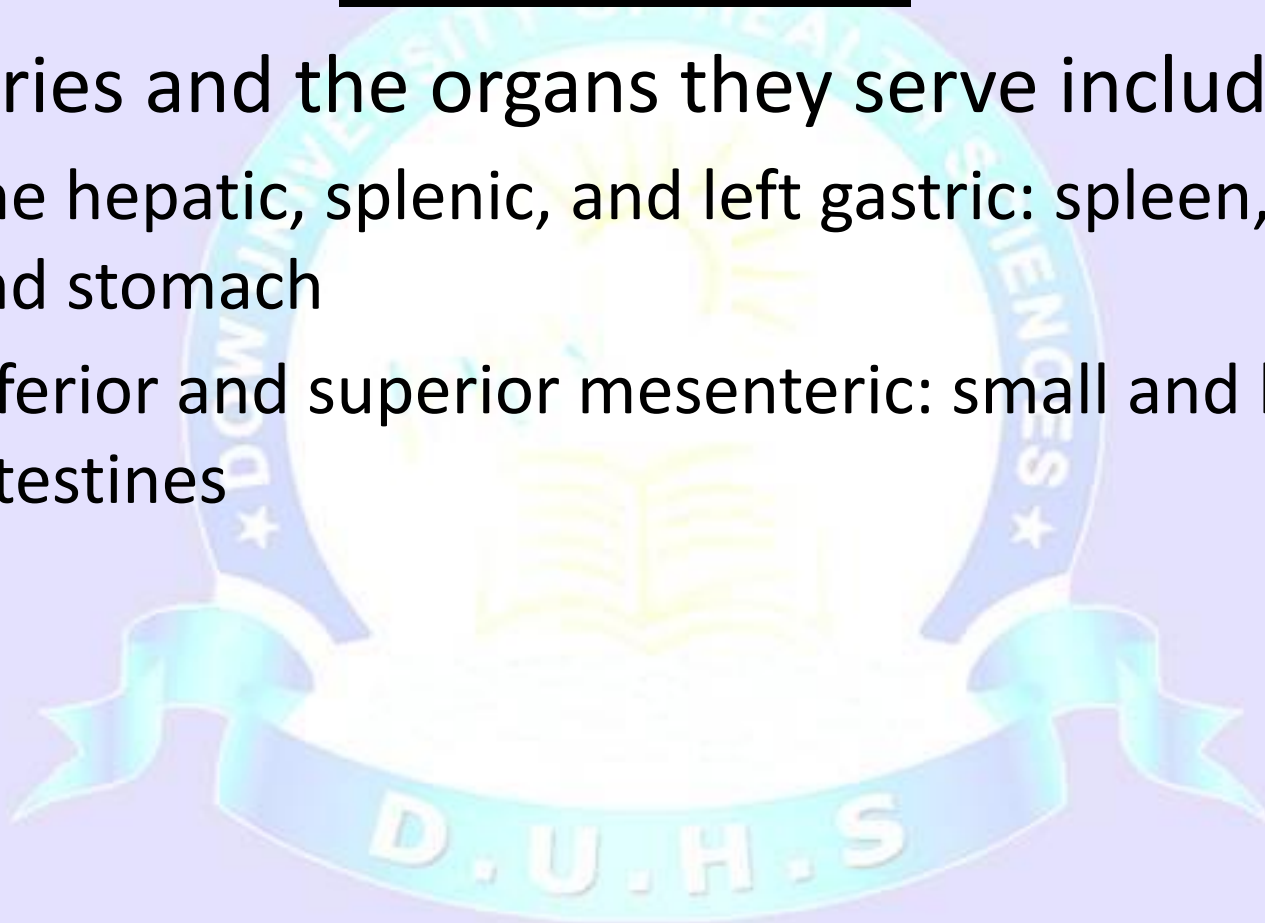


Figure 23.4

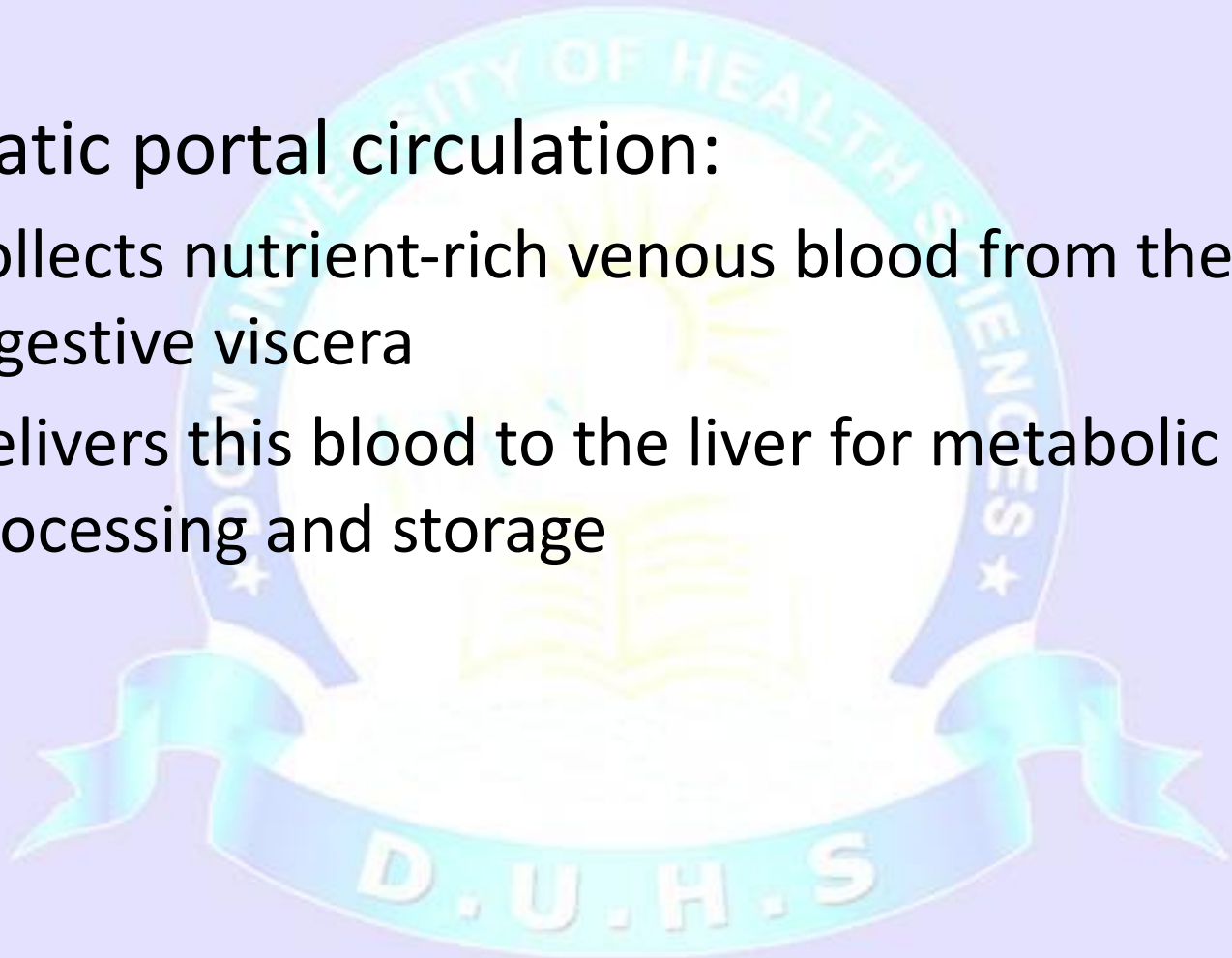
Blood Supply: Splanchnic Circulation

- Arteries and the organs they serve include
 - The hepatic, splenic, and left gastric: spleen, liver, and stomach
 - Inferior and superior mesenteric: small and large intestines



Blood Supply: Splanchnic Circulation

- Hepatic portal circulation:
 - Collects nutrient-rich venous blood from the digestive viscera
 - Delivers this blood to the liver for metabolic processing and storage



Blood supply to GI system

Blood Supply: Splanchnic Circulation

- Celiac trunk (hepatic, splenic, L. gastric) to spleen, liver, & stomach
- sup & inf mesenteric to sm & lg intestines
 - Celiac trunk & mesenterics receive $\frac{1}{4}$ of C.O.
 - percentage increases after ingestion of a meal
- hepatic portal system collects nutrient rich venous blood draining from digestive viscera & takes it to the liver
 - liver collects nutrients for metabolism or storage b/f releasing them back to the bloodstream (via hepatic vein) for general cellular use

Thanks!

